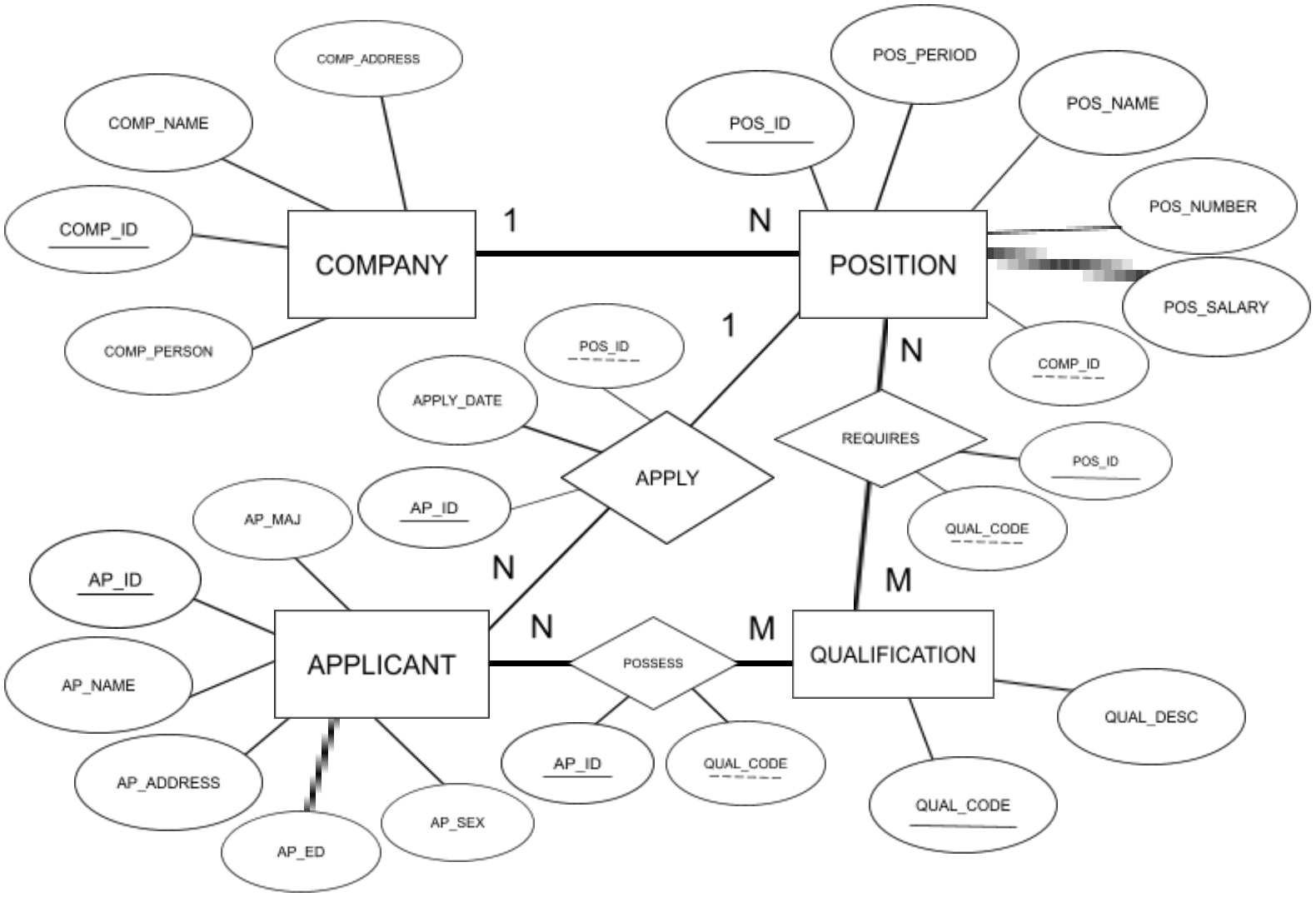
กิจกรรมที่ 1


COMP_ADDRESS
COMP_NAME
COMP_ID
COMP_PERSON
COMPANY
1
N
POS_PERIOD
POS_ID
POS_NAME
POS_NUMBER
POSITION
POS_SALARY
COMP_ID
1
N
POS_ID
APPLY_DATE
APPLY
REQUIRES
POS_ID
QUAL_CODE
AP_MAJ
AP_ID
AP_ID
N
M
N
M
APPLICANT
POSSESS
QUALIFICATION
AP_NAME
QUAL_DESC
AP_ADDRESS
AP_ID
QUAL_CODE
QUAL_CODE
AP_SEX
AP_ED

กิจกรรมที่ 2

COMPANY

COMP_ID	COMP_NAME	COMP_ADDRESS	COMP_PERSON

POSITION

POS_ID	POS_NAME	POS_PREIOD	POS_SALARY	POS_NUMBER	COMP_ID

APPLICANT

AP_ID	AP_NAME	AP_ADDRESS	AP_SEX	AP_ED	AP_MAJ

QUALIFICATION

QUAL_CODE	QUAL_DESC

POS_QUAL

POS_ID	QUAL_CODE

APPLY

AP_ID	POS_ID	APPLY_DATE

AP_QUAL

AP_ID	QUAL_CODE

กิจกรรมที่ 3

COMPANY

COMP_ID	COMP_NAME	COMP_ADDR	COMP_CHW	COMP_PERSON
_____				

POSITION

POS_ID	POS_NAME	POS_PREIOD	POS_SALARY	POS_NUMBER	COMP_ID
_____					_ _ _ _ _ _ _

APPLICANT

AP_ID	AP_NAME	AP_ADDR	AP_CHW	AP_SEX	AP_ED	AP_MAJ
_____						

QUALIFICATION

QUAL_CODE	QUAL_DESC
_____	

POS_QUAL

POS_ID	QUAL_CODE
_____	_ _ _ _ _ _ _

APPLY

AP_ID	POS_ID	APPLY_DATE
_____	_ _ _ _ _ _ _	

AP_QUAL

AP_ID	QUAL_CODE
_____	_ _ _ _ _ _ _

กิจกรรมที่ 4

---

1)

COMPANY

```
CREATE TABLE company (
  COMP_ID CHAR(4) NOT NULL PRIMARY KEY,
  COMP_NAME VARCHAR(15) NOT NULL,
  COMP_ADDR VARCHAR(30) NOT NULL,
  COMP_CHW CHAR(30) NOT NULL,
  COMP_PERSON CHAR(20) NOT NULL
);
```

POSITION

```
CREATE TABLE position (
  POS_ID CHAR(5) NOT NULL PRIMARY KEY,
  POS_NAME VARCHAR(25) NOT NULL,
  POS_PERIOD CHAR(2) NOT NULL,
  POS_SALARY DECIMAL(20, 2) NOT NULL,
  POS_NUMBER INTEGER NOT NULL,
  COMP_ID CHAR(4) NOT NULL REFERENCES company(COMP_ID)
);
```

APPLICANT

```
CREATE TABLE applicant (
  AP_ID CHAR(4) NOT NULL PRIMARY KEY,
  AP_NAME VARCHAR(30) NOT NULL,
  AP_ADDR VARCHAR(50),
  AP_CHW CHAR(50),
  AP_SEX CHAR(1),
  AP_ED CHAR(30),
  AP_MAJ CHAR(5)
);
```

QUALIFICATION

```
CREATE TABLE qualification (
  QUAL_CODE CHAR(10) NOT NULL PRIMARY KEY,
  QUAL_DESC VARCHAR(50)
);
```

POS_QUAL

```
CREATE TABLE pos_qual (
  POS_ID CHAR(5) NOT NULL REFERENCES position(POS_ID),
  QUAL_CODE CHAR(10) NOT NULL REFERENCES qualification(QUAL_CODE),
  PRIMARY KEY (POS_ID, QUAL_CODE)
);
```

APPLY

```
CREATE TABLE apply (
  AP_ID CHAR(4) NOT NULL REFERENCES applicant(AP_ID),
  POS_ID CHAR(5) NOT NULL REFERENCES position(POS_ID),
  APPLY_DATE DATE,
  PRIMARY KEY (AP_ID, POS_ID)
);
```

AP_QUAL

```
CREATE TABLE ap_qual (
  AP_ID CHAR(4) NOT NULL REFERENCES applicant(AP_ID),
  POS_ID CHAR(5) NOT NULL REFERENCES position(POS_ID),
  PRIMARY KEY (AP_ID, POS_ID)
);
```

2)

2.1) เพิ่มคอลัมน์ COMP_PHONE

```
ALTER TABLE company
ADD COMP_PHONE CHAR(10);
```

2.2 เปลี่ยนประเภทข้อมูล COMP_PHONE เป็น NUMERIC

```
ALTER TABLE company
ALTER COLUMN COMP_PHONE NUMERIC(10);
```

2.3 ลบคอลัมน์ COMP_PHONE

```
ALTER TABLE company
DROP COLUMN COMP_PHONE;
```

2.4 เปลี่ยนขนาด COMP_NAME เป็น 20 ตัวอักษร

```
ALTER TABLE company
ALTER COLUMN COMP_NAME CHAR(20);
```

3)

COMPANY

```
INSERT INTO company VALUES ('C001', 'google', 'USA, 101/10','CA','supanut');
INSERT INTO company VALUES ('C002', 'yumyum', 'TH','BKK', 'boy');
INSERT INTO company VALUES ('C003', 'BetaSoft', 'Thailand, BKK', 'Somchai T.');
```

POSITION

```
INSERT INTO position VALUES ('PS001', 'Technical Programmer', '12', 25000.00, 2, 'C001');
INSERT INTO position VALUES ('PS002', 'Web Developer', '6', 30000.00, 1, 'C002');
INSERT INTO position VALUES ('PS010', 'ICT Manager', '12', 22000.00, 3, 'C003');
```

APPLICANT

```
INSERT INTO applicant VALUES ('A001', 'Napat', '101 m.2 bangna', 'Bangkok', 'M', 'Bachelor',
'math',);
INSERT INTO applicant VALUES ('A002', 'Suda', '71/102 m.4', 'Chiang Mai', 'F', 'Bachelor', 'comsci');
INSERT INTO applicant VALUES ('A003', 'Anan', '1/1 m.1', 'Phuket', 'M', 'Master', 'IT');
```

QUALIFICATION

```
INSERT INTO qualification VALUES ('ASP', 'Web_ASP');
INSERT INTO qualification VALUES ('C++', 'C++');
INSERT INTO qualification VALUES ('DB2', 'Database Administrator,DB2');
```

POS_QUAL

```
INSERT INTO pos_qual VALUES ('PS001', 'ASP');
INSERT INTO pos_qual VALUES ('PS002', 'C++');
INSERT INTO pos_qual VALUES ('PS003', 'DB2');
```

APPLY

```
INSERT INTO apply VALUES ('A001', 'PS001', '2025-01-01');
INSERT INTO apply VALUES ('A002', 'PS002', '2025-02-03');
INSERT INTO apply VALUES ('A003', 'PS003', '2025-03-05');
```

AP_QUAL

```
INSERT INTO apply VALUES ('A001', 'PS001');
INSERT INTO apply VALUES ('A002', 'PS002');
INSERT INTO apply VALUES ('A003', 'PS003');
```

4)

```
DROP TABLE company;
```

กิจกรรมที่ 5

1.DELETE FROM qualification WHERE QUAL_CODE = ‘C++’

2.UPDATE company SET COMP_NAME = ‘นายบุญมี สุขมาก’ WHERE COMP_NAME = ‘BetaSoft’;

3.UPDATE position SET POS_NUMBER = 4 WHERE POS_NAME = ‘ICT Manager’;

4.SELECT * FROM company COMP_ID DESC;

5.SELECT AP_ID, AP_NAME, AP_SEX, AP_MAJ FROM applicant WHERE (AP_NAME LIKE 'A%' OR AP_NAME LIKE '%Y%');

6.SELECT MAX(POS_SALARY) AS MaxSalary,MIN(POS_SALARY) AS MinSalary, AVG(POS_SALARY) AS AvgSalary,COUNT(POS_NAME) AS PosNumCount FROM position;

7.SELECT DISTINCT AP_CHW FROM applicant WHERE AP_SEX = ‘M’;

8.SELECT POS_ID, POS_NAME, POS_SALARY,POS_PERIOD FROM position WHERE POS_PERIOD BETWEEN 1 AND 3 AND POS_SALARY >= 20000;

9.SELECT POS_MAJ, COUNT(*) AS total_ap FROM applicant GROUP BYPOS_MAJ;

10.SELECT c.COMP_ID, c.COMP_NAME, p.POS_NAME, p.POS_NUMBER FROM company c, position p WHERE c.COMP_ID = p.COMP_ID;